

Multi-modal and multi-attribute generation of single cells with CFGEN

Palma et al., bioRxiv, 2025
2025.03.12 Journal Meeting



KOREA
UNIVERSITY

Theis lab

ML in Single-Cell Genomics



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single-cell-tutorialPublic

Single cell current best practices tutorial case study for the paper: Lunken and Theis, "Current best practices in single-cell RNA-seq analysis: a tutorial"

Jupyter Notebook1.5k470

single-cell-best-practicesPublic

https://www.sc-best-practices.org

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scveloPublic

RNA Velocity generalized through dynamical modeling

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CellRank: dynamics from multi-view single-cell data

Python37248

scarchesPublic

Reference mapping for single-cell genomics

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Benchmarking analysis of data integration tools

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Machine LearningDeep LearningComputational Biology

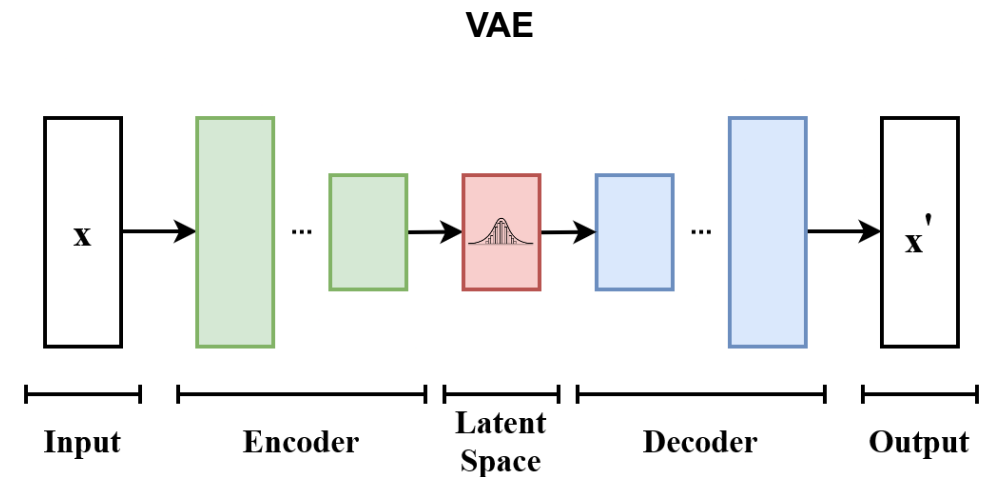
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TITLE	CITED BY	YEAR
Predicting cell morphological responses to perturbations using generative modeling A Palma, FJ Theis, M Lotfollahi Nature Communications 16 (1), 505	14*	2025
Generating Multi-Modal and Multi-Attribute Single-Cell Counts with CFGen A Palma, T Richter, H Zhang, M Lubetzki, A Tong, A Dittadi, F Theis The Thirteenth International Conference on Learning Representations	1	2024
Mixed Models with Multiple Instance Learning JP Engelmann*, A Palma*, JM Tomczak, F Theis, FP Casale International Conference on Artificial Intelligence and Statistics, 3664-3672	7*	2024
cellFlow: a generative flow-based model for single-cell count data A Palma, T Richter, H Zhang, A Dittadi, F Theis ICLR 2024 MLGenX workshop	1*	2024

Related Works

Deep learning models of single cell omics

- With the advent of deep generative models, **VAE**-based approaches have proven remarkably flexible for batch correction(scVI), Integration btw several modality(totalVI)
- These methods focus on learning meaningful **cellular representations** rather than **generating synthetic datasets** from noise
- Insteads, tasks involving realistic cell generation have been explored using other models such as diffusion models (e.g., scDiffusion), GANs (e.g., scGAN), and LLMs
 - *cf. scDiffusion utilizes Gaussian diffusion with log-normalized counts*
- however, they rely on **continuous** representations, which are not fully aligned with the inherently **discrete** nature of single-cell count data
- **VAE models**, which can integrate discrete statistical distribution could be a breakthrough of this issue



Why do we need the “Raw” counts?

Generating the realistic data from CFGen

1. Incorporating **discrete noise** into generative models enhances the effectiveness of batch effect correction
 2. For raw count-based analyses (e.g., Batch correction, DEG), preserving data discreteness is essential
- Due to the incompatibility of discrete data with continuous models such as Gaussian diffusion, **most approaches generate data pre-processed through normalization and scaling**
 - **CFGen, utilizes VAE model samples latent variables probabilistically, allowing integration with a statistical distribution(e.g NB distribution) to effectively model the discrete count data**

Backgrounds

Backgrounds

Probabilistic Representation of Single-cell Gene Expression and scATAC Data

- Gene expression and DNA accessibility is represented as **high-dim. vectors of discrete counts**
 - Representing **Sparsity, Over-dispersion**
- So that, we use Negative-Binomial distribution to model the expression counts in single cell data
- assume that each cell has an individual mean and the gene-wise over-dispersion
- DNA accessibility is measured by the presence(1) or absence(0) of a signal peak as DNA region accessibility
- By its binary characteristic, it follows the Bernoulli model
- CFGen utilizes these **DISCRETE noise models for each data modality** that represents the original data distributions as above

[The expr. of gene g in cell n]

$$x_{ng} \sim \text{NB}(\mu_{ng}, \theta_g)$$

$$\text{Gene variance} = \mu + \frac{\mu^2}{\theta}$$

[The # of accessible DNA region(peaks) p in cell n]

$$b_{np} \sim \text{Bernoulli}(\pi_{np})$$

where π_{np} = cell – gene specific success probability

Backgrounds

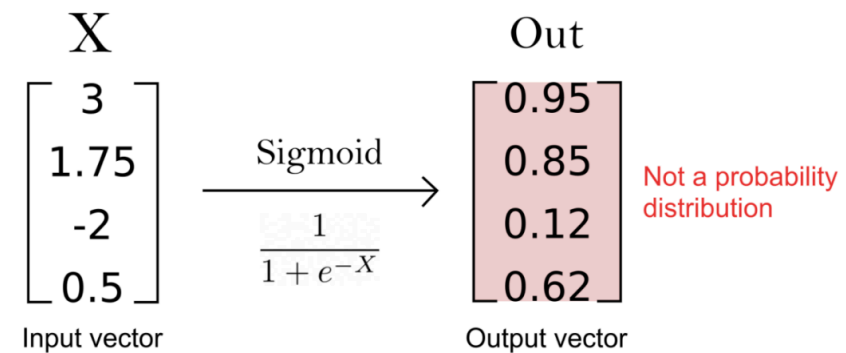
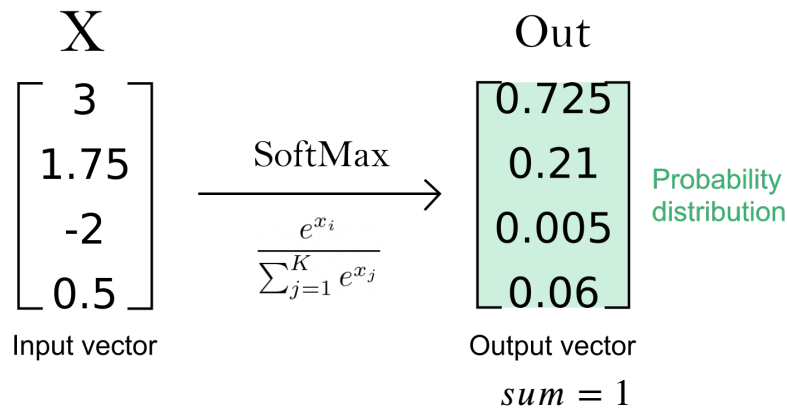
Decoder Parameterization for Single-cell Expression and Accessibility

a given latent variable

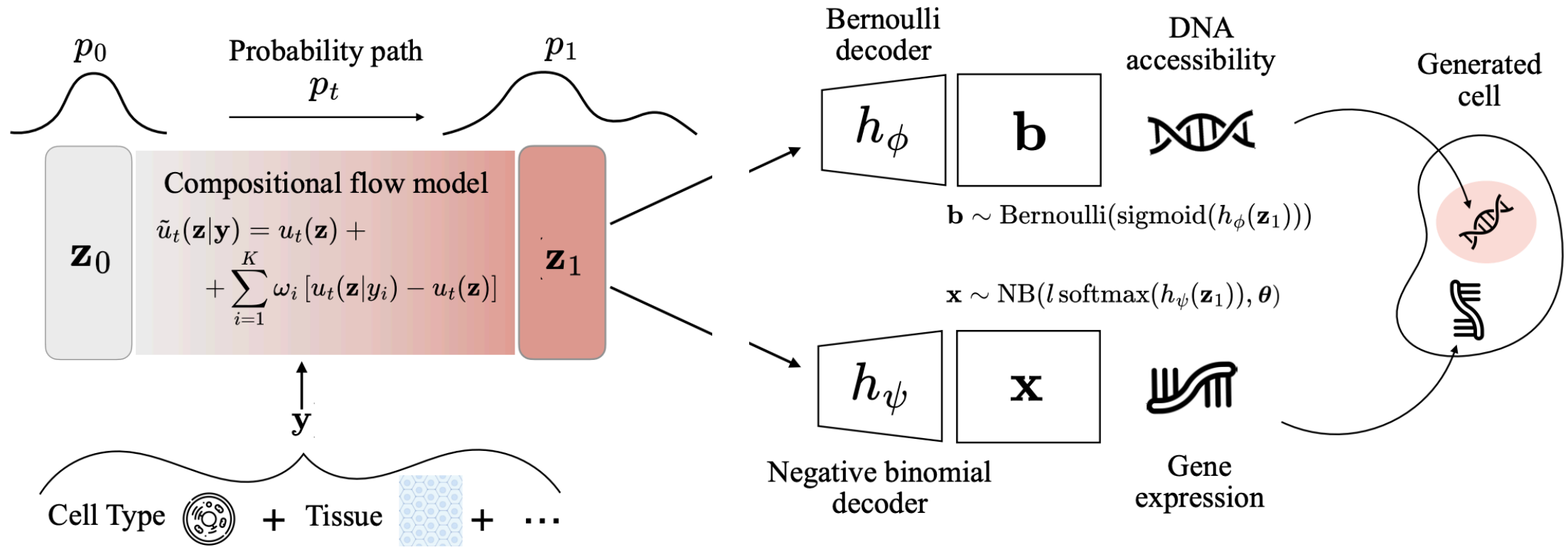
$$\mu = l\rho, \quad \rho = \text{softmax}(h_\psi(\mathbf{z})), \quad \pi = \text{sigmoid}(h_\phi(\mathbf{z}))$$

where h_ψ, h_ϕ are the modality – specific decoders
and l is the size factor (total counts in each cell)

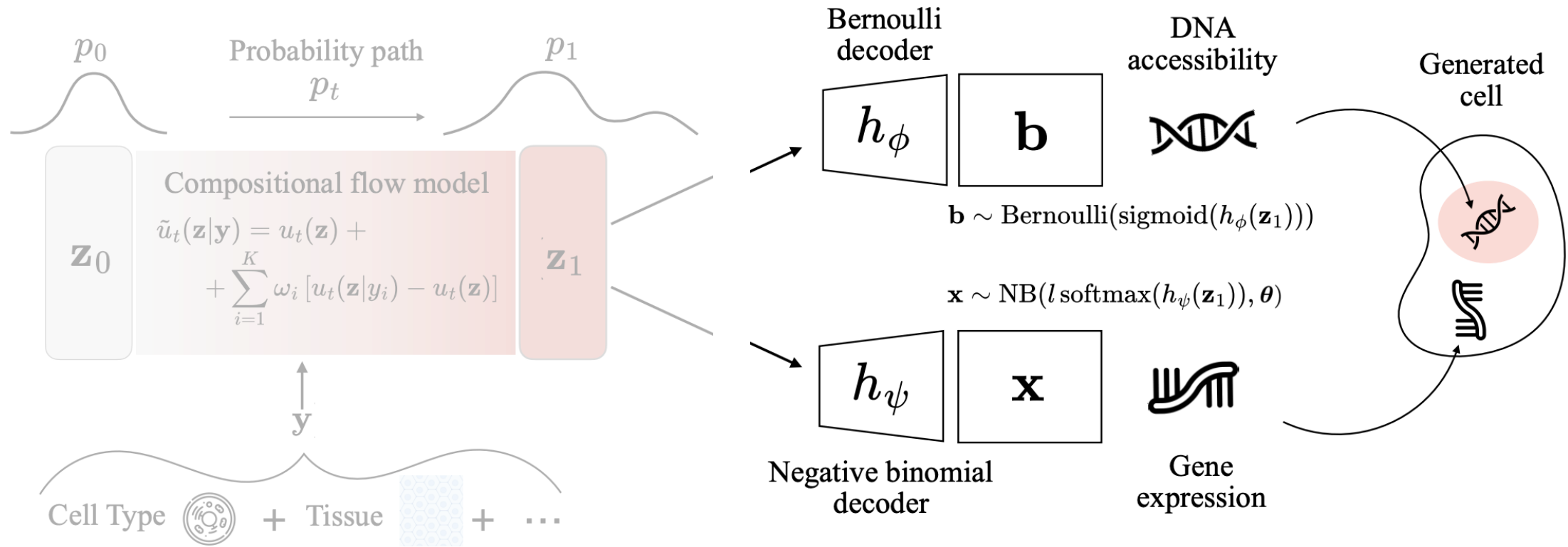
- Activation Func.



Generating single-cell data with CFGen

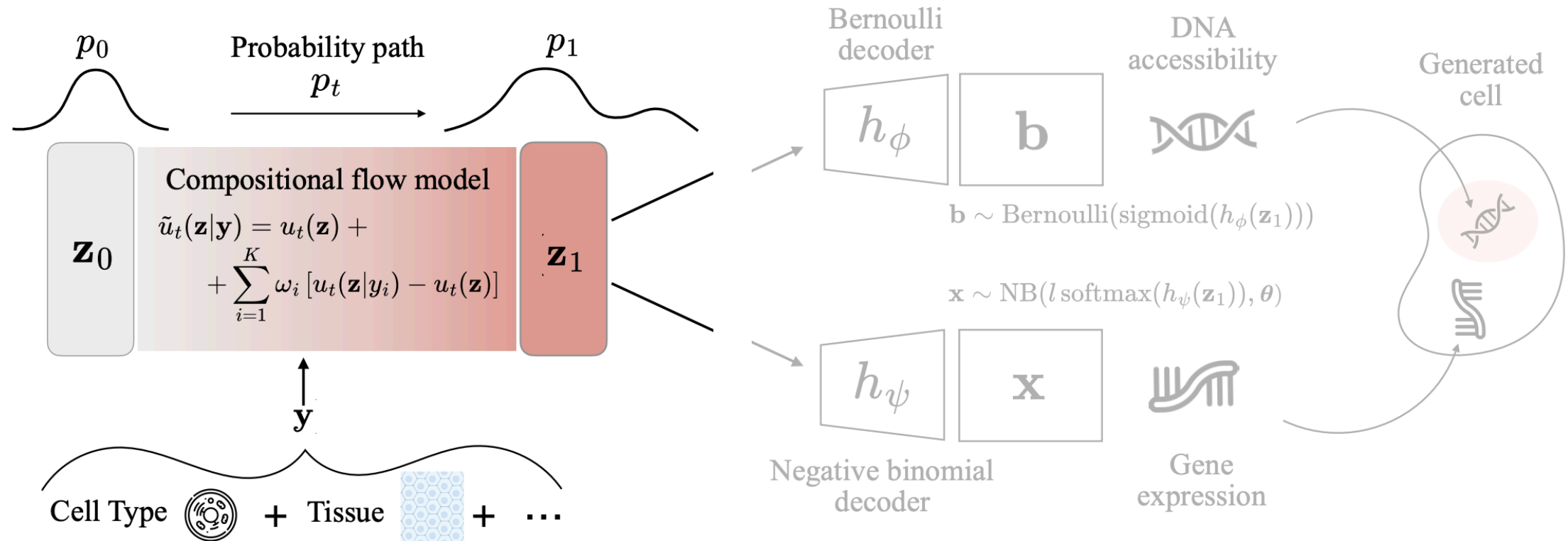


Generating single-cell data with CFGen



- Generating "Discrete" single cell counts / Accessibility from a latent variable z_1

Generating single-cell data with CFGen

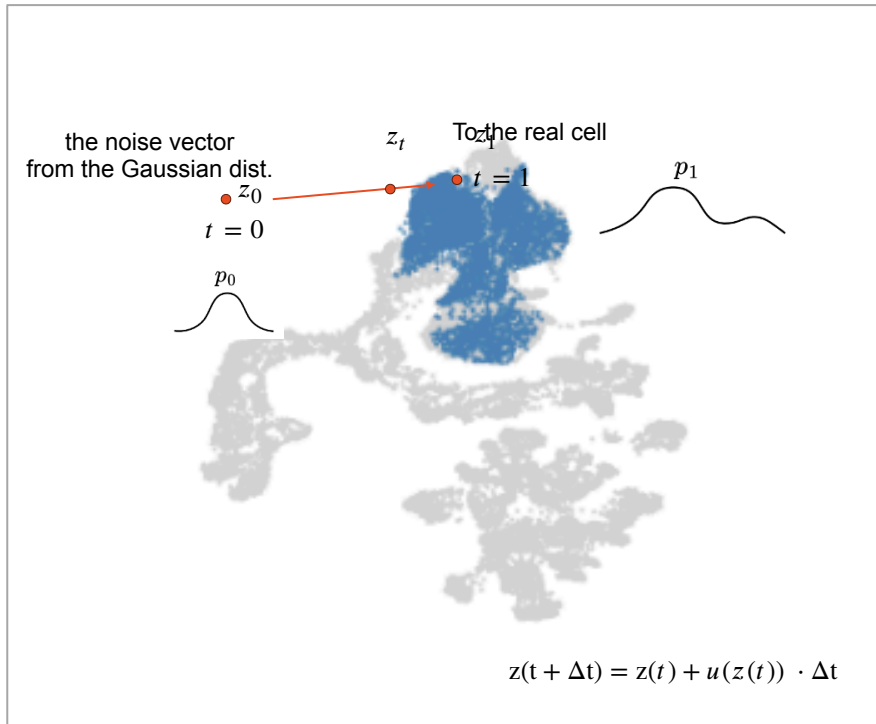


- Continuous Normalizing Flows(CNF)
- **Flow Matching**
- **Classifier-Free Guidance(CFG)**

The algorithm of Multi-Attribute Flow Matching

Training the velocity field neural network with CFGen model

$v_{t,\xi}$



Algorithm 1 Train CFGen with multiple attributes on scRNA-seq

Require: Probability of unconditional generation p_{uncond} , trained encoder f_η , scheduling (α_t, σ_t) .

```

1: Initialize  $v_{t,\xi}$ 
2: while not converged do
3:   Sample  $(\mathbf{x}, y_1, \dots, y_K)$  from the data
4:    $\mathbf{z}_1 \leftarrow f_\eta(\mathbf{x})$ 
5:   Sample  $t$  from  $\mathcal{U}[0, 1]$ 
6:    $l \leftarrow$  Sum of entries of  $\mathbf{x}$ 
7:   Sample  $b$  from Bernoulli( $p_{\text{uncond}}$ )
8:   if  $b = 1$  then
9:      $y \leftarrow \emptyset$ 
10:  else
11:     $y \leftarrow$  sample uniformly a label among  $y_1, \dots, y_K$ 
12:  end if
13:   $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  {sample noise}
14:   $\mathbf{z}_t \leftarrow \alpha_t \mathbf{z}_1 + \sigma_t \mathbf{z}_0$  {noisy data point}
15:   $\dot{\mathbf{z}}_t \leftarrow \dot{\alpha}_t \mathbf{z}_1 + \dot{\sigma}_t \mathbf{z}_0$ 
16:  Take gradient step on  $\nabla_\xi ||v_{t,\xi}(\mathbf{z}_t, y, l) - \dot{\mathbf{z}}_t||^2$ 
17: end while

```

Encoding x into z_1

Sampling t

$\dot{\mathbf{z}}_t$ as a suppl. Datapoint to cal. loss

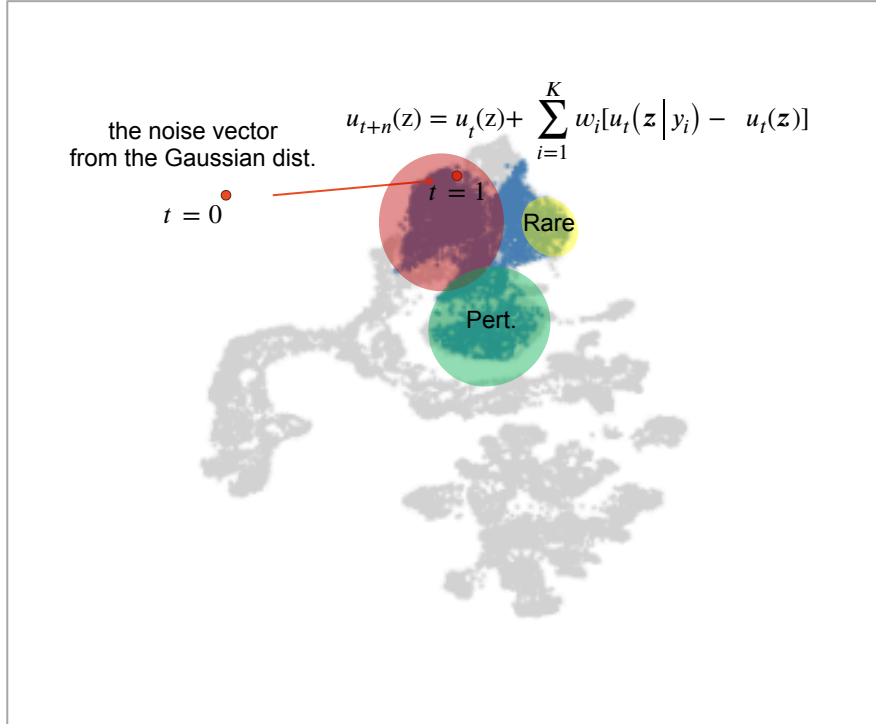
Calculating the optimal vector field which minimize the loss

Output: $v_{t,\xi}$

- f_η : Encoder based on VAE
- p_t : Probability path, describing the change of data distribution in t
- $v_{t,\xi}$: the vector field neural network

The algorithm of Multi-Attribute Flow Matching

Guiding data generation on a specific condition with CFGen model



Algorithm 1 Train CFGen with multiple attributes on scRNA-seq

Require: Probability of unconditional generation p_{uncond} , trained encoder f_η , scheduling (α_t, σ_t) .

```

1: Initialize  $v_{t,\xi}$ 
2: while not converged do
3:   Sample  $(\mathbf{x}, y_1, \dots, y_K)$  from the data
4:    $\mathbf{z}_1 \leftarrow f_\eta(\mathbf{x})$ 
5:   Sample  $t$  from  $\mathcal{U}[0, 1]$ 
6:    $l \leftarrow$  Sum of entries of  $\mathbf{x}$ 
7:   Sample  $b$  from  $\text{Bernoulli}(p_{\text{uncond}})$ 
8:   if  $b = 1$  then
9:      $y \leftarrow \emptyset$ 
10:  else
11:     $y \leftarrow$  sample uniformly a label among  $y_1, \dots, y_K$ 
12:  end if
13:   $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  {sample noise}
14:   $\mathbf{z}_t \leftarrow \alpha_t \mathbf{z}_1 + \sigma_t \mathbf{z}_0$  {noisy data point}
15:   $\dot{\mathbf{z}}_t \leftarrow \dot{\alpha}_t \mathbf{z}_1 + \dot{\sigma}_t \mathbf{z}_0$ 
16:  Take gradient step on  $\nabla_\xi ||v_{t,\xi}(\mathbf{z}_t, y, l) - \dot{\mathbf{z}}_t||^2$ 
17: end while

```

Unconditional

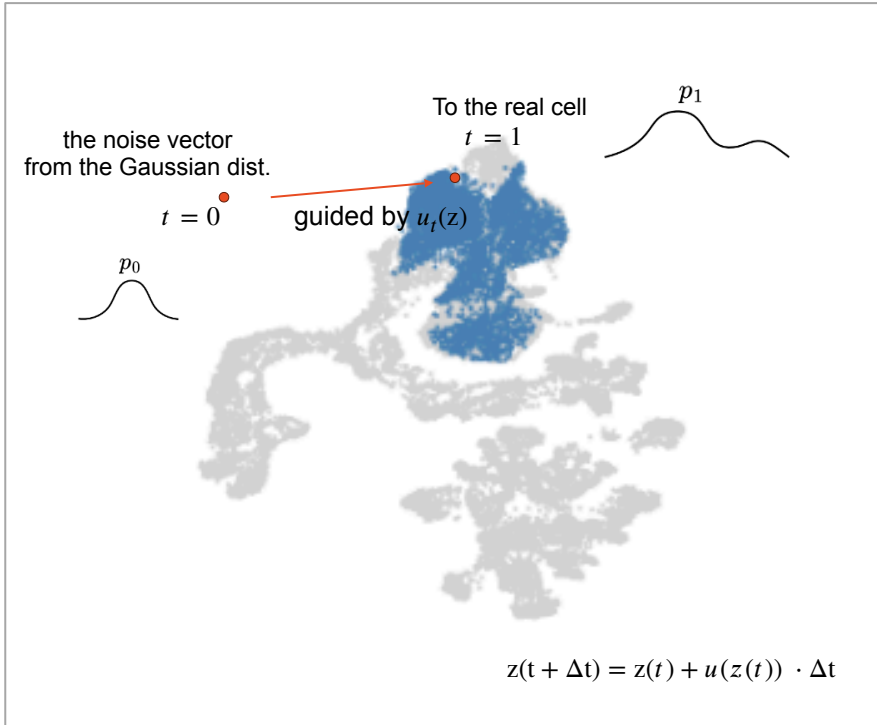
Conditional

- w : a guidance strength hyperparameter for each condition y
- y : a condition(attribute)

The algorithm of Multi-Attribute Flow Matching

Sampling from the multi-attribute trained model

$v_{t,\xi}$



Algorithm 2 Sampling from multi-attribute guided CFGen for scRNA-seq

Require: Trained velocity field $v_{t,\xi}$, conditions $y_1, \dots, y_K$, guidance parameters $\omega_1, \dots, \omega_K$, size factor distribution parameters (μ_l, σ_l) , number of ODE steps n_{ode} , trained decoder h_ψ , trained inverse dispersion parameter θ .

- 1: Sample size factor l from $\text{LogNormal}(\mu_l, \sigma_l)$
- 2: $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ {sample noise} Sampling noise vector $\mathbf{z}_0$
- 3: $n \leftarrow 1/n_{\text{ode}}$ {step size}
- 4: $\tilde{u}_t(\cdot) \leftarrow v_{t,\xi}(\cdot, \emptyset, l) + \sum_{i=1}^K \omega_i [v_{t,\xi}(\cdot, y_i, l) - v_{t,\xi}(\cdot, \emptyset, l)]$ {guided velocity function} Unconditional
Conditional
- 5: **for** $t = 0, n, \dots, 1 - n$ **do**
- 6: $\mathbf{z}_{t+n} \leftarrow \text{ODEStep}(\tilde{u}_t, \mathbf{z}_t)$ {ODE solver step} $\mathbf{z}_t \rightarrow \mathbf{z}_{t+n}$ by vector field
- 7: **end for**
- 8: $\mathbf{x} \leftarrow \text{Sample from NB}(l \text{ softmax}(h_\psi(\mathbf{z}_1)), \theta)$ Decoding $\mathbf{x}$ from $\mathbf{z}$

→ A vector field with multiple attribute

Output: $\mathbf{x}$

- $u_t(z)$: the marginal vector field with the time-conditioned neural network $v_{t,\xi}$
- Sampling from compositions of attributes is obtained by integrating the parametrized field $\tilde{v}_{t,\xi}(z, y)$ starting from a Gaussian prior p_0

Experiments

- Benchmarking & Application -

Benchmarking

Quantitative performance comparison

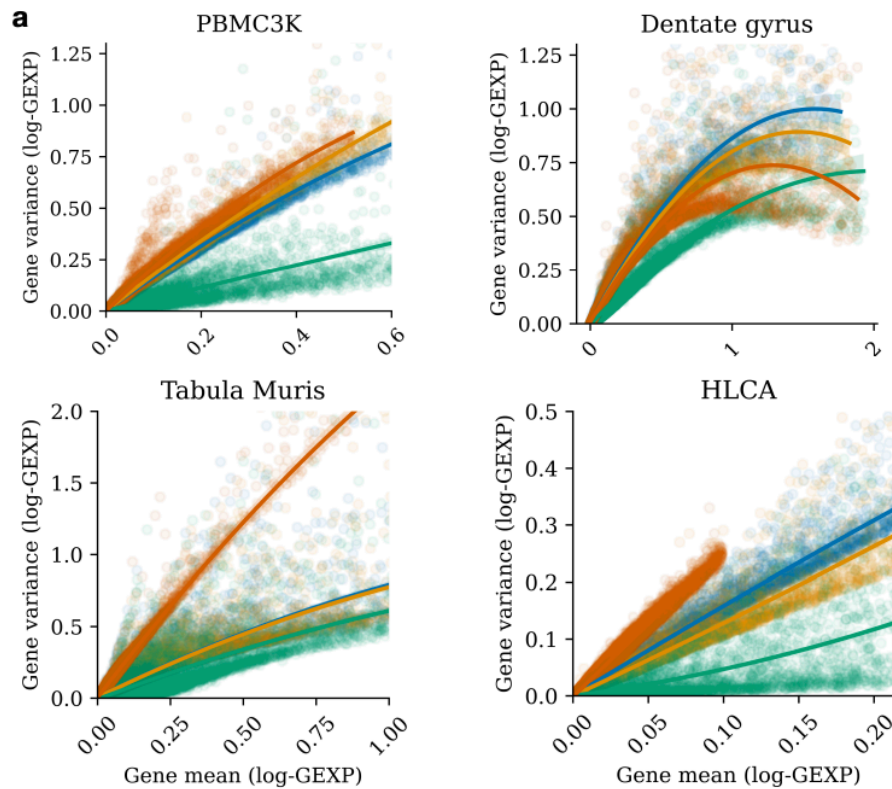
- **Evaluation metrics**
 - **MMD**(RBF-kernel Maximum Mean Discrepancy) : Distance btw real and augmented data distribution using Kernal function
 - **2-Wasserstein distance** : The cost of aligning the distribution when the distributions are different using optimal transport
- Since scDiffusion and scGAN generate normalized data while CFGen and scVI produce discrete counts, the discrete counts **were log-normalized after the augmentation**

	Counts: 2,638		18,213		245,389		584,944	
	PBMC3K		Dentate gyrus		Tabula Muris		HLCA	
	MMD (↓)	WD (↓)	MMD (↓)	WD (↓)	MMD (↓)	WD (↓)	MMD (↓)	WD (↓)
Conditional								
c-CFGen	0.85 ± 0.05	16.94 ± 0.44	1.12 ± 0.04	21.55 ± 0.17	0.19 ± 0.02	7.39 ± 0.20	0.54 ± 0.02	10.72 ± 0.08
scDiffusion	1.27 ± 0.20	22.41 ± 1.21	1.22 ± 0.05	22.56 ± 0.10	0.24 ± 0.04	7.89 ± 0.45	0.96 ± 0.04	15.82 ± 0.45
scVI	0.94 ± 0.05	17.66 ± 0.29	1.15 ± 0.04	22.61 ± 0.23	0.26 ± 0.02	9.76 ± 0.53	0.58 ± 0.02	11.78 ± 0.19
Unconditional								
u-CFGen	0.44 ± 0.01	16.81 ± 0.06	0.42 ± 0.01	21.20 ± 0.02	0.08 ± 0.00	8.54 ± 0.06	0.15 ± 0.01	10.63 ± 0.01
scGAN	0.36 ± 0.01	15.54 ± 0.06	0.42 ± 0.01	22.52 ± 0.03	0.25 ± 0.00	12.85 ± 0.04	0.18 ± 0.01	10.81 ± 0.01

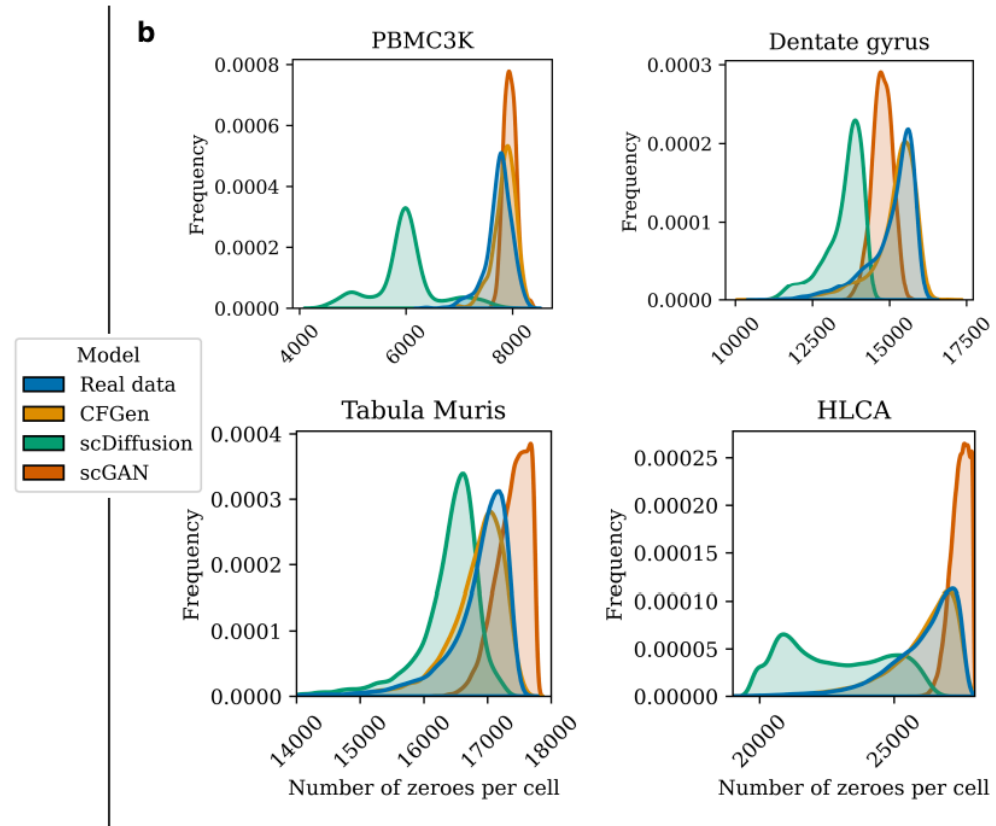
Benchmarking

Comparison of distribution matching metrics

- Over-dispersion

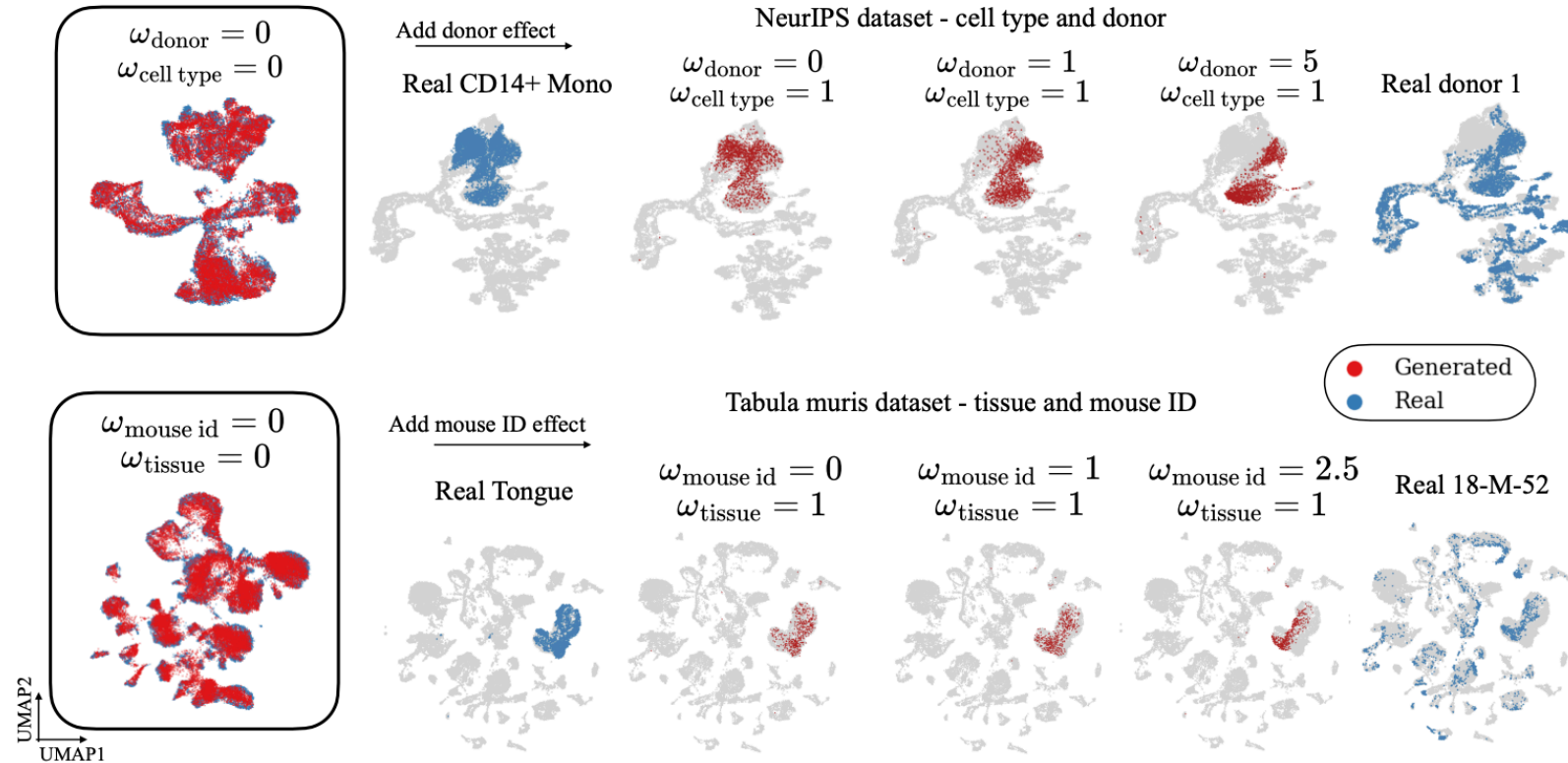


- Sparsity



Qualitative comparison of the generation results

Qualitative evaluation of guidance performance on attribute pairs



- Increasing the guidance weight steers the generation to the intersection of the two attributes

Application 1.

Data augmentation to improve classification of rare cell types

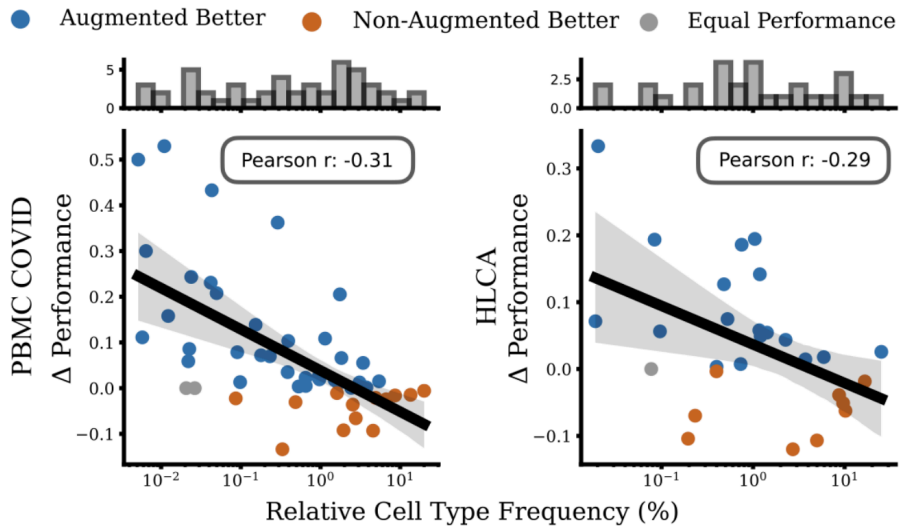


Table 8: Table reporting the cell type classification recall of scGPT before (Recall Base) and after (Recall Aug) augmentation on the HLCA dataset. The Relative Frequency (%) column reports how rare a certain cell type is in the dataset. For each row, we highlight which setting leads to the highest recall.

Cell Type	Relative Frequency (%)	Recall Base	Recall Aug
alveolar macrophage	20.00	0.95	0.95
tracheobronchial serous cell	0.05	0.43	0.64
myofibroblast cell	0.04	0.25	0.69
conventional dendritic cell	0.04	0.58	0.81
mucus secreting cell	0.03	0.61	0.61
dendritic cell	0.02	0.46	0.70
mesothelial cell	0.02	0.91	1.00
smooth muscle cell	0.02	0.09	0.15
lung neuroendocrine cell	0.02	0.97	0.97
brush cell of tracheobronchial tree	0.01	0.21	0.37
stromal cell	0.01	0.35	0.88
fibroblast	0.01	0.30	0.60
hematopoietic stem cell	0.01	0.78	0.89
tracheobronchial goblet cell	0.01	0.00	0.50

- HLCA: 584,944 cells → 800,000 cells
- PBMC: 422,220 cells → 800,000 cells
- Upsampling rare cell types to the inverse of their cell type frequency
- *However, still did not reach uniformity, as this imbalance may be biologically meaningful*
- Fitting a kNN cell-type classifier on the pretrained scGPT embeddings from the original and augmented training sets and evaluating the recall performance on held-out set
- suggests **the additional synthetic examples help the model better distinguish the cell type** in the LLM representation space

Application 2.

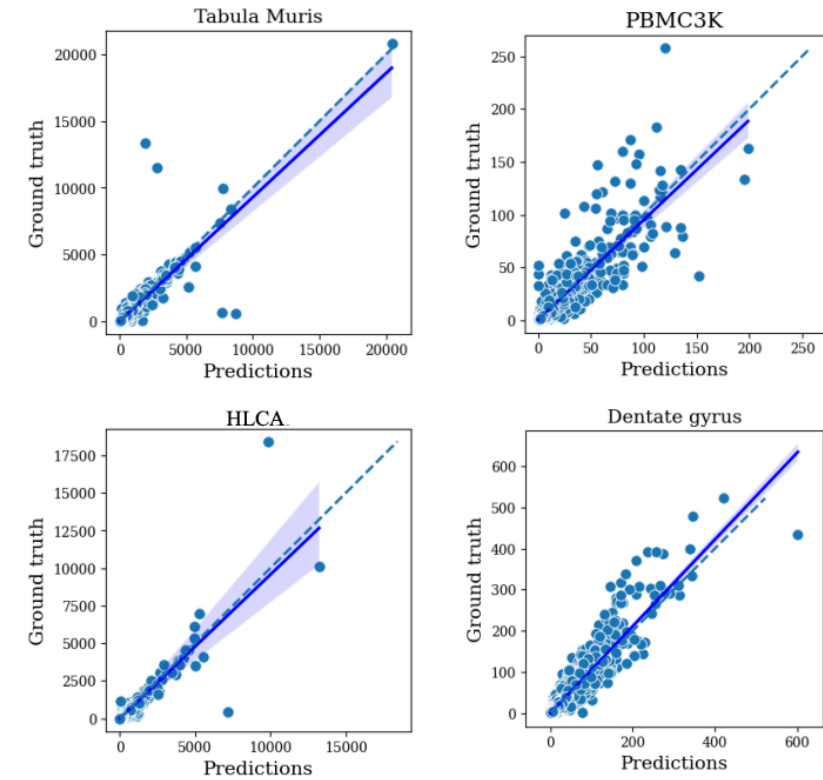
Missing gene imputation with CFGen

- **Inverting the generative flow** to compute the data into noise(z) and then **re-generating new gene expression** including imputed gene expression
- *While scVI imputates the missing gene expr. with the probabilistic approaches in latent space*
- An imputation strategy using CFGen showed a better gene expression inference performance compared to scVI with the 10% masking genes.
 - L1 distance(MAE, Manhattan distance)
 - $\sum |\text{Expr. Real} - \text{Expr. Imputed}|$

Dataset name	Number of cells	Number of genes	Number of cell types
PMBC3K	2,638	8,573	8
Dentate gyrus	18,213	17,002	14
Tabula Muris	245,389	19,734	123
HLCA	584,944	27,997	50
PBMC10k	10,025	25,604	14
NeurIPS	90,261	14,087	45
PBMC COVID	422,220	2,000	29
C.Elegans	89,701	17,747	35 (plus unknown)

Table 9: Mean distance and correlation between real and imputed genes by scVI and CFGen.

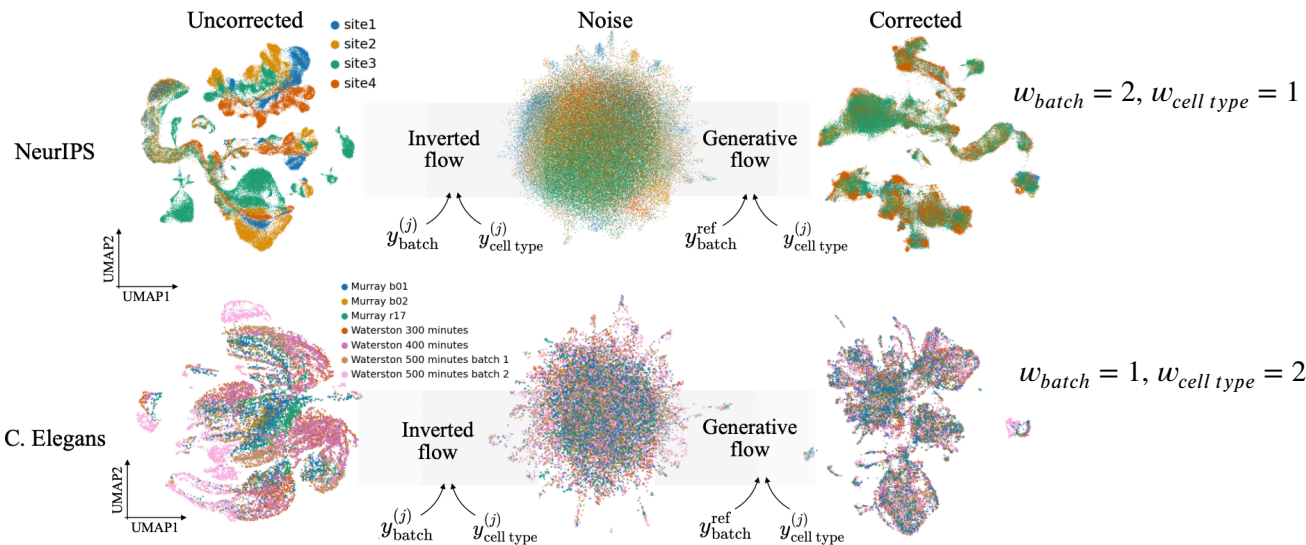
	Mean L ₁ distance real-imputed counts (↓)				Pearson correlation real-imputed counts (↑)			
	PBMC3K	Dentate gyrus	HLCA	T. Muris	PBMC3K	Dentate gyrus	HLCA	T. Muris
CFGen	1.21	0.42	3.21	4.81	0.68	0.56	0.83	0.86
scVI	1.47	0.35	4.43	6.08	0.61	0.58	0.75	0.79



Application 3.

Batch correction

- Workflow for batch correction with CFGen
 - Flow inversion** to remove both biological, technical variation from $z \rightarrow$ Encoding expression x into a latent variable z with the attributes $y_{batch}, y_{cell\ type}$
 - Generating new observations by simulation of **the flow forward again** preserving only biological attribute by controlling the weights $w_{batch}, w_{cell\ type}$

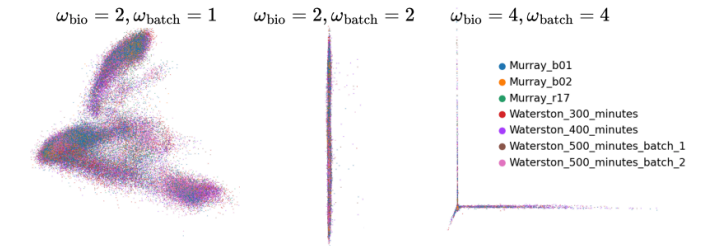


- Excessive guidance appear optimal as they got higher scIB metric scores,
- but might cause an unnatural collapse of variability beyond the batch and biological annotation, leading to **artefacts**
- Recommending** $w_{bio}, w_{batch} \in \{1,2\}$ **After observing the severity of batch effects manually**

scIB pkg

	NeurIPS		C. Elegans	
	Batch (↑)	Bio (↑)	Batch (↑)	Bio (↑)
CFGen	0.63	0.61	0.68	0.63
scPoli	0.55	0.64	0.61	0.56
scANVI	0.48	0.68	0.61	0.59
scVI	0.45	0.63	0.58	0.55
Uncorrected	0.33	0.62	0.40	0.53

Qualitative evaluation of guidance weights



Quantitative evaluation of guidance weights

Guidance weights	C. Elegans	
	Batch Correction	Bio Conservation
$w_{bio}=0\ w_{batch}=0$	0.48	0.55
$w_{bio}=1\ w_{batch}=1$	0.67	0.55
$w_{bio}=1\ w_{batch}=2$	0.68	0.54
$w_{bio}=2\ w_{batch}=1$	0.68	0.63
$w_{bio}=2\ w_{batch}=2$	0.69	0.63
$w_{bio}=2\ w_{batch}=3$	0.69	0.64
$w_{bio}=3\ w_{batch}=2$	0.70	0.67
$w_{bio}=3\ w_{batch}=3$	0.70	0.68
$w_{bio}=3\ w_{batch}=4$	0.70	0.67
$w_{bio}=4\ w_{batch}=4$	0.70	0.69

Discussion

Discussion

- **Scalability**
 - Data Augmentation
 - Conditional Generation
- **Limitation**
 - Assuming the independence in the data → **Ignoring co-expression**
 - **Data distribution imbalance** remains a crucial challenge in AIVC construction